

Butterfly Survey

Swadini, Blyde River Canyon



Mpumalanga · South Africa

Seasonal observations 2019 – 2025

1,959

TOTAL RECORDS

170

SPECIES

7

SEASONS

12

CORE SPECIES

Observer

Andries de Vries · Citizen Science · Observation.org

1. Introduction & Study Context

This report summarises seven consecutive seasons of butterfly (Lepidoptera) observations made by citizen scientist Andries de Vries at Swadini, a Forever Resort situated within the Blyderivierspoort Nature Reserve, Mpumalanga, South Africa. The study area is defined by a User Location created on Observation.org, covering the western bank of the Blyde River below the Blyderivierspoort Dam, within the Blyde River Canyon itself and also within the larger Blyde River Canyon Nature Reserve. The area covers approximately 1.1 km² and is characterised by proximity to the river, indigenous forest, and forest-margin habitat. Observations by other visitors are also recorded on Observation.org but are excluded from this dataset; the platform does not permit downloading other observers' records for privacy reasons.

Observations were originally submitted to the Virtual Museum at the University of Cape Town (vmus.adu.org.za) and subsequently transferred to the international citizen science platform Observation.org. The dataset used for this analysis was exported from Observation.org in September 2025 and contains the observer's own butterfly records for the defined Swadini User Location.

Surveys were conducted annually during extended camping stays of approximately four months, though the timing, duration, and intensity of effort varied between seasons as summarised in Table 1 below.

Table 1 — Season Summary

Year	Records	Species	IOC Weeks	Notes
2019	98	57	18 – 31 / 46 – 47	First butterfly season; primary focus on Odonata. Two visits: May 2 – Aug 4 (wks 18–31) and Nov 11–18 (wks 46–47). 2 records on 30 Sep 2019 excluded (incorrectly attributed; observed at Izinkwe Game Lodge, Limpopo).
2020	81	48	33 – 42	Lockdown delayed arrival to July; shorter window
2021	398	83	16 – 35	Increased focus on Lepidoptera; effort markedly higher
2022	356	79	17 – 34	Consistent effort; comfortable with families and genera
2023	413	85	17 – 33	Peak season richness; good conditions throughout
2024	196	58	17 – 44	Severe drought; insect and bird numbers notably low
2025	415	92	18 – 35	Highest species count; 3-week gap in July (family emergency)

Species counts are normalised to species level — subspecies records have been rolled up to their parent species to ensure comparability across years.

2. Data & Methods

Normalisation. The raw dataset contains records classified as species, subspecies, aggregate species, and synonyms. For all analyses in this report, subspecies records were normalised by truncating the scientific name to genus and species (i.e. the third name component was removed). This yielded 170 unique species across all seven seasons.

IOC Week numbers. Dates were converted to ISO 8601 week numbers (Monday-start, weeks 1–52/53). This system is used throughout as the primary temporal unit because it allows direct comparison of the same calendar period across different years, independent of the day of the week on which a season

started.

Comparable weeks. Because survey windows differed between years, a subset of "comparable weeks" was identified: IOC weeks that were surveyed in at least 5 of the 7 seasons (i.e. $\geq 75\%$ year coverage). Eleven weeks qualified: weeks 18–22, 24–26, 29, 31, and 33. Analyses using this subset are explicitly noted.

Validation. Of the 1,959 records, 1,719 carry "accepted (with evidence)" status, 31 were accepted by automatic validation, and 209 remain with "unknown" status — primarily 2025 observations not yet reviewed by a validator. The observer applied a personal confidence threshold of approximately 95% for photo-free records submitted in 2025.

3. Species Richness

A total of **170 unique species** were recorded across all seven seasons (1,957 records after correction of 2 misattributed 2019 records). Annual species richness ranged from 48 (2020) to 92 (2025). The annual average across the five high-effort seasons (2021–2025) is 79 species, compared to which the observer's cumulative total is almost double — reflecting the value of multi-year presence at a single site.

The drought year 2024 stands out sharply: despite being the second-longest survey window (weeks 17–44) it produced only 58 species and 196 records — comparable to 2019 when the observer was still primarily focused on Odonata. This strongly suggests that the low species count in 2024 reflects genuine ecological scarcity rather than observer effort.

2025 achieved the highest single-season species count (92) despite a three-week gap in July and a survey window largely confined to autumn and early winter. This may partly reflect improved recording of species without photographic evidence, as well as genuinely good post-drought recovery conditions.



Left: *Phalanta phalantha aethiopica* — African Leopard (Nymphalidae) · Right: *Colotis ione* — Bushveld Purple Tip (Pieridae) · Swadini, Blyde River Canyon · Photos: Andries de Vries

Table 2 — Species per Family per Year

Family	2019	2020	2021	2022	2023	2024	2025
Hesperiidae	5	3	12	11	12	9	13
Lycaenidae	11	6	18	18	17	9	21
Nymphalidae	30	28	37	34	40	27	41
Papilionidae	5	4	8	8	8	7	9
Pieridae	7	7	8	8	8	6	8

Nymphalidae consistently dominate, comprising 46–56% of species each year. The 2024 drought reduced every family, but *Lycaenidae* (gossamer-winged butterflies) showed the steepest proportional decline — from 17–18 species in 2021–2023 to only 9 in 2024.

4. Core Resident Species

Twelve species were recorded in all seven seasons, regardless of survey duration, observer effort, or climatic conditions. These are considered the core resident butterfly fauna of the Swadini study area:

Species	Common Name	Family
<i>Belenois creona</i>	African Common White	Pieridae
<i>Bicyclus anynana</i>	Squinting Bush Brown	Nymphalidae
<i>Byblia ilithyia</i>	Spotted Joker	Nymphalidae
<i>Charaxes phaeus</i>	Demon Emperor	Nymphalidae
<i>Colotis evagore</i>	Small Orange Tip	Pieridae
<i>Hemiolaus caeculus</i>	Tailed Blue	Lycaenidae
<i>Junonia hierta</i>	Yellow Pansy	Nymphalidae
<i>Junonia natalica</i>	Natal Pansy	Nymphalidae
<i>Melanitis leda</i>	Common Evening Brown	Nymphalidae
<i>Neptis saclava</i>	Banded Sailer	Nymphalidae
<i>Pseudacraea lucretia</i>	False Wanderer	Nymphalidae
<i>Tagiades flesus</i>	Clouded Forester	Hesperiidae



Charaxes saturnus saturnus — Foxy Charaxes
(Nymphalidae)

The *Charaxes* genus — emperors and commodores — are among the most spectacular butterflies at Swadini. Several *Charaxes* species feature in the dataset, including *C. brutus*, *C. xipharex*, and *C. phaeus*; the latter is one of the 12 core resident species recorded in all seven seasons.

Photo: Andries de Vries · Swadini · September 2020

Nine of the twelve core species belong to Nymphalidae. Notably, only one Lycaenidae species (*Hemiolaus caeculus*) qualifies, suggesting that gossamer-winged butterflies at Swadini are more sensitive to annual variation in conditions. The presence of all twelve even during the drought year 2024 and the truncated lockdown season 2020 confirms their status as genuinely resident rather than seasonal migrants.

5. Temporal Patterns — IOC Week Analysis

Because the survey window differed between years, direct week-by-week comparison is only valid for weeks that were covered in most seasons. With the corrected 2019 dataset, sixteen comparable weeks were identified (surveyed in ≥ 5 of 7 years): weeks 18–33. Table 3 presents species counts for these weeks.

Precis tugela tugela — Dead-leaf Commodore (Nymphalidae)

Precis tugela is one of the 12 core resident species recorded in all seven seasons at Swadini. Together with three *Junonia* species (*J. hierta*, *J. natalica*, *J. oenone*, *J. terea*), this group forms the most consistently present assemblage in the dataset. Their year-round presence makes them reliable indicators of general butterfly activity levels across seasons.

Photo: Andries de Vries · Swadini · August 2021



Table 3 — Species per Comparable IOC Week

Week	2019	2021	2022	2023	2024	2025	Avg	Max
Wk 18	12	12	6	7	12	8	9.5	12
Wk 19	6	9	10	16	7	34	13.7	34
Wk 20	9	2	10	20	31	12	14.0	31
Wk 21	9	17	3	23	18	20	15.0	23
Wk 22	13	17	10	41	2	39	20.3	41
Wk 23	6	14	17	21	13	22	15.5	22
Wk 24	2	20	9	18	25	17	15.2	25
Wk 25	6	6	8	29	3	31	13.8	31
Wk 26	3	3	14	17	7	29	12.2	29
Wk 27	1	16	19	21	11	22	15.0	22
Wk 28	4	21	21	20	8	23	16.2	23
Wk 29	7	27	8	2	5	13	10.3	27
Wk 30	2	10	8	12	6	20	9.7	20
Wk 31	5	14	18	1	5	9	8.7	18
Wk 32	5	10	14	12	10	20	11.8	20
Wk 33	.	17	32	15	2	20	17.2	32

2023 column shown in green (high richness); 2024 column shown in red (drought year). A dot (.) indicates the week fell outside the survey window that year (2020 excluded as it covered only one comparable week).

Week 22 (late May/early June) shows the greatest variability: 41 species in 2023 and 39 in 2025, against only 2 in 2024. This single week encapsulates the impact of the 2024 drought more starkly than any other metric. Weeks 19–22 collectively represent the peak activity period across most seasons, corresponding to autumn in the Southern Hemisphere.

6. Species Absences & Notable Patterns

Of the 170 species recorded, 91 were absent in at least one season despite being recorded in two or more other seasons. The core question driving this research is whether these absences reflect genuine ecological absence (habitat change, drought, population fluctuation) or are artefacts of limited survey effort.

Effort-related absences. Absences in 2020 (short lockdown window, weeks 33–42 only) and to a lesser extent 2019 (low effort, Odonata focus) should be interpreted cautiously. Species recorded only in spring or early summer would be entirely missed by a survey beginning in week 33.

Drought-related absences. The 2024 season provides the clearest signal of ecologically meaningful absence. The combination of normal survey timing (weeks 17–44), extended coverage, but dramatically reduced species count (58 species, 196 records) is consistent with the observer's field impression of low insect and bird numbers throughout the season. Lycaenidae showed the sharpest decline.

Species of particular interest. Several species were regularly recorded in 2021–2023 but absent in 2024 despite adequate survey coverage: *Bicyclus anynana*, *Papilio ophidicephalus*, *Charaxes brutus*, *Charaxes xiphares*, and multiple *Acraea* species. Their return in 2025 suggests the absences were drought-driven rather than indicative of permanent local decline.

7. Notes on Methodology & Limitations

Survey effort varies. Records per season ranged from 81 to 415. Species richness correlates partly with effort, so direct year-to-year comparison of totals should be treated as indicative rather than strictly quantitative.

Habitat uniformity assumed. All records fall within ~1.1 km². Habitat heterogeneity (riverine, forest, margin) exists but proximity makes formal habitat partitioning inconclusive.

Unvalidated 2025 records. Approximately 20% of 2025 records carry "unknown" validation status. These are included in all analyses; the observer applied a ~95% personal confidence threshold for photo-free submissions.

Phenology. The fixed survey window (approximately weeks 17–35 for most years) does not permit determination of full annual phenology. Week numbers serve as a relative temporal index within the season only.

Subspecies normalisation. Rolling subspecies up to species level is a deliberate simplification. In a small number of cases, records flagged as different subspecies of the same species from different years may obscure taxonomic change.

8. Interactive Dashboard & AI Attribution

An interactive HTML dashboard accompanies this report. It contains six chart sections: species richness by year, observation effort and IOC week coverage, family breakdown, a week-by-year species diversity heatmap, a filterable presence/absence matrix grouped by family, an individual species IOC week chart, and the comparable-weeks analysis described in Section 5. The dashboard is a self-contained HTML file that opens in any modern web browser without installation.

AI Attribution

This summary report and the accompanying interactive HTML dashboard were generated by **Claude**, an AI assistant developed by **Anthropic** (claude.ai · anthropic.com), at the request of the observer in April 2026.

The analysis, charts, and written interpretation were produced by Claude based on the observation data exported from Observation.org by the observer. All biological interpretation, field knowledge, and contextual notes were provided by the observer. Claude performed data processing, normalisation, statistical summaries, visualisation code, and report writing.

The data remains the intellectual property of the observer and Observation.org. This report is intended for sharing with fellow lepidopterists and site managers as a summary of citizen science activity at Swadini.

Report generated: April 2026 · Data export: September 2025 · Total records analysed: 1,957 (corrected) · Species (normalised): 170